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DEPARTMENT OF SOFTWARE ENGINEERING

COURSE: OPERATING Systems and Systems Programming

SECTION-A

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SUBMISSION DATE: Apr 28, 2026 G.C.

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1. Installation of Qubes

1.1. Introduction

Operating systems form the fundamental layer of software that manages computer hardware and software resources, providing common services for computer programs. They are essential for the functioning of any digital device, from personal computers to complex servers. Among the diverse landscape of operating systems, **Qubes OS** stands out due to its unique approach to security. Developed with a strong focus on compartmentalization, Qubes OS aims to provide a secure computing environment by isolating different applications and tasks into separate virtual machines.

Qubes OS achieves this isolation by leveraging the **Xen hypervisor**, a Type-1 bare-metal hypervisor that runs directly on the hardware. This architecture allows Qubes OS to create multiple, isolated virtual environments, known as "qubes" or AppVMs, each running a different operating system (E.g: Fedora, Debian, Windows) or a specific application. This design significantly enhances security by preventing a compromise in one qube from affecting others or the entire system.

The motivation for choosing Qubes OS for this assignment stems from its innovative security model and its reliance on advanced virtualization concepts. Understanding its architecture provides valuable insights into modern operating system design and security principles. However, this assignment also presents a deliberate challenge: attempting to install Qubes OS within **VMware Workstation**, a Type-2 hypervisor. This approach is known to have significant limitations and technical challenges due to Qubes OS's inherent design for bare-metal execution with Xen. This exercise will serve as a practical exploration of nested virtualization complexities and the specific incompatibilities between Qubes OS and VMware, offering a deeper learning experience regarding hypervisor differences and system requirements.

1.2. Objectives

General Objective

To gain a comprehensive understanding of operating system installation within a virtualized environment, specifically focusing on the unique challenges and architectural considerations of Qubes OS when deployed using VMware Workstation.

Specific Objectives

- To document the step-by-step process of attempting Qubes OS installation within VMware Workstation.
- To identify and thoroughly explain the technical limitations and issues encountered during the installation, particularly those arising from the incompatibility between Qubes OS's Xen hypervisor and VMware's virtualization layer.
- To analyze and compare the filesystem support in Qubes OS with other common filesystems, understanding their implications for security and performance.
- To evaluate the advantages and disadvantages of Qubes OS, especially in the context of its security model and system requirements.
- To propose realistic solutions and recommendations for successful Qubes OS deployment, considering the observed limitations.

1.3. Requirements

1.3.1 Hardware Requirements

For a successful Qubes OS installation, even in a simulated environment, the host machine must meet specific hardware criteria:

- **CPU**: A modern multi-core processor (Intel VT-x/EPT or AMD-V/RVI) with hardware virtualization extensions enabled in the BIOS/UEFI firmware. Qubes OS heavily relies on these features for isolation.
- **RAM**: A minimum of 8GB of RAM is required for basic functionality, with 16GB or more highly recommended for optimal performance, especially when running multiple qubes concurrently.
- **Storage**: A Solid State Drive (SSD) is strongly recommended due to the intensive disk I/O operations performed by Qubes OS. A minimum of 128GB of free space is advisable.
- **BIOS/UEFI**: Hardware virtualization (Intel VT-x/AMD-V) and Intel VT-d/AMD-Vi (IOMMU) must be enabled in the system's firmware settings. IOMMU is crucial for device pass-through and isolation in Qubes OS.

1.3.2 Software Requirements

- **VMware Workstation**: The virtualization software (E.g: VMware Workstation Pro) used to create and manage the virtual machine.
- **Qubes OS ISO file**: The installation image for Qubes OS (E.g: Qubes-R4.2.0-x86_64.iso).
- **Host Operating System**: A stable host operating system (E.g: Windows, Linux, macOS) capable of running VMware Workstation and supporting nested virtualization.

1.4. Installation Steps (Attempted in VMware Workstation)

1.4.1 Downloading Qubes OS ISO

Step 1: Visit Official Website

- Open browser
- Go to the official Qubes OS website

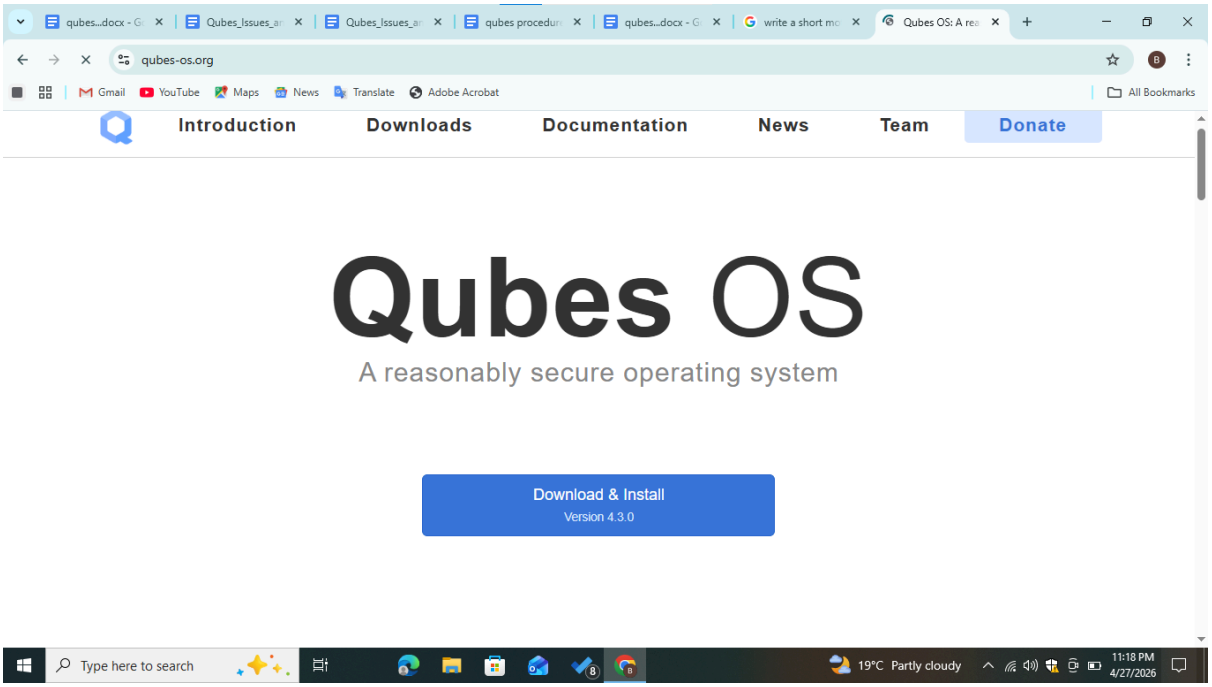

Figure 1: qubes-os.org interface

Step 2: Navigate to Downloads

- Click on Download Qubes OS

Step 3: Select Latest Version

- Choose the latest stable release(4.3)

Step 4: Download ISO File

- Click the ISO download link

1.4.2 Installing VMware Workstation Pro

Step 1: Download VMware

- Visit VMware official website
- Click Download Now

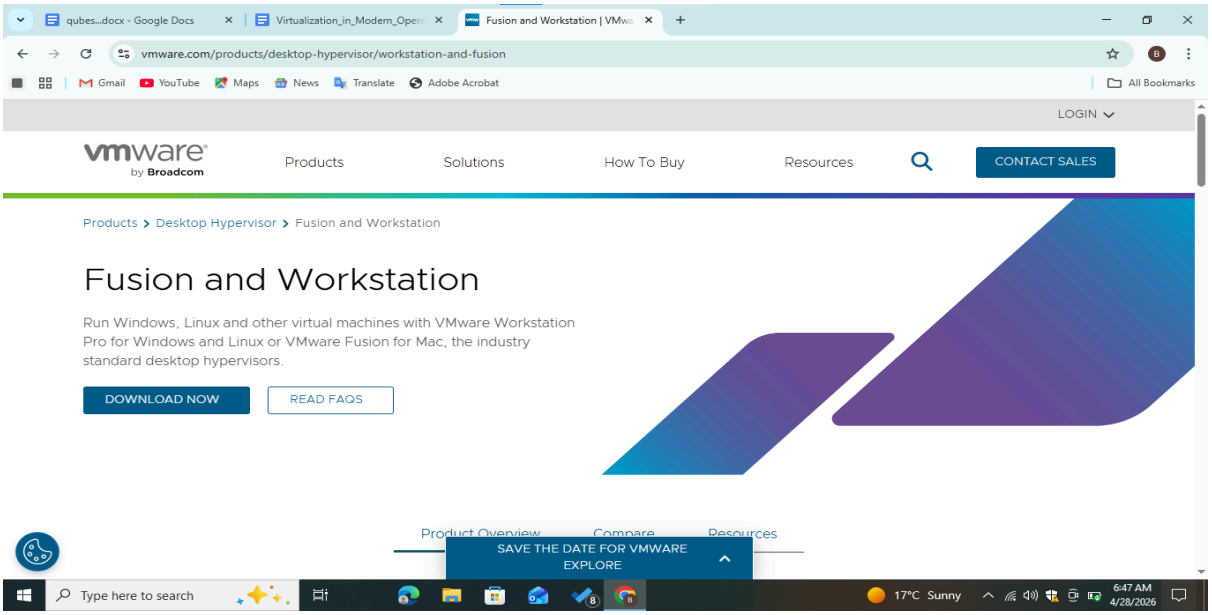

Figure 2: vmware.com interface

Step 2: sign in to broadcom

- ➢ Navigate to the My Downloads section on the Broadcom dashboard
- ➢ Access the free software downloads area to find VMware products
- ➢ Complete the trade compliance and additional verification forms

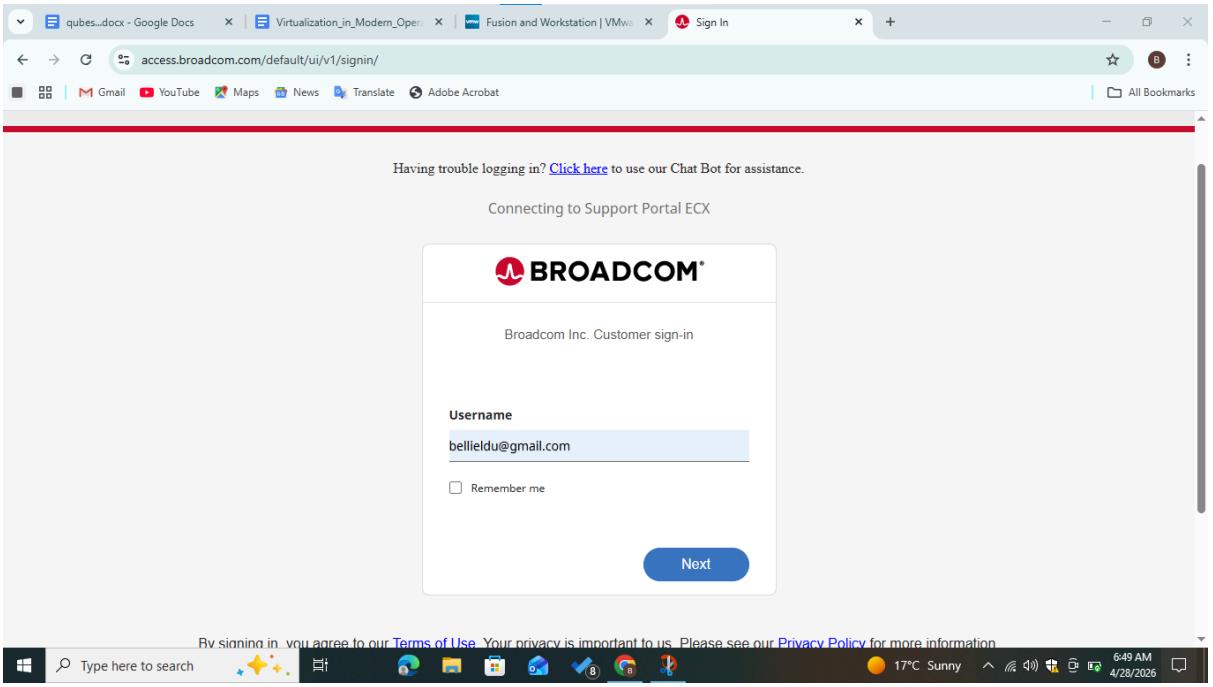

Figure 3: broadcom.com interface

Step 3: System Preparation and Optimization

A. Disabling Conflicting Windows Features

Certain Windows features must be disabled to allow VMware to use native CPU virtualization

- ➢ Open the Turn Windows features on or off menu in the Control Panel
- ➢ Uncheck Windows Hyper-V and Virtual Machine Platform
- ➢ Uncheck Windows Hypervisor Platform and Windows Sandbox
- ➢ Uncheck Windows Subsystem for Linux (WSL)
- ➢ Apply changes and restart the computer to complete the removal of these features

B. Security and BIOS Settings

- ➢ Disable Memory Integrity in Windows Security
  - ➔ Navigate to Device Security then Core isolation details
  - ➔ Toggle the Memory integrity switch to the off position and restart if prompted
- ➢ Verify CPU Virtualization is enabled in the system BIOS/UEFI
  - ➔ Check status in Task Manager under the Performance tab
  - ➔ If disabled, enter BIOS and enable Intel VT-x (intel) or AMD-V/SVM (AMD)
  - ➔ This is required for running 64-bit guest operating systems with high performance

Step 4: Run Installer

- Launch the downloaded VMware Workstation Pro installer
- Follow the setup wizard prompts
  - ➔ Accept the End User License Agreement
  - ➔ The installer will verify that Hyper-V is not detected to ensure native performance
- ➢ Choose installation preferences such as shortcuts and product update checks
- ➢ Click Install to begin the process and Finish once the installation is complete

1.4.3 Creating Virtual Machine

Step 1: Launch VMware

- Open VMware Workstation Pro

Step 2: Create New VM

- Select Create New Virtual Machine
- Choose Typical

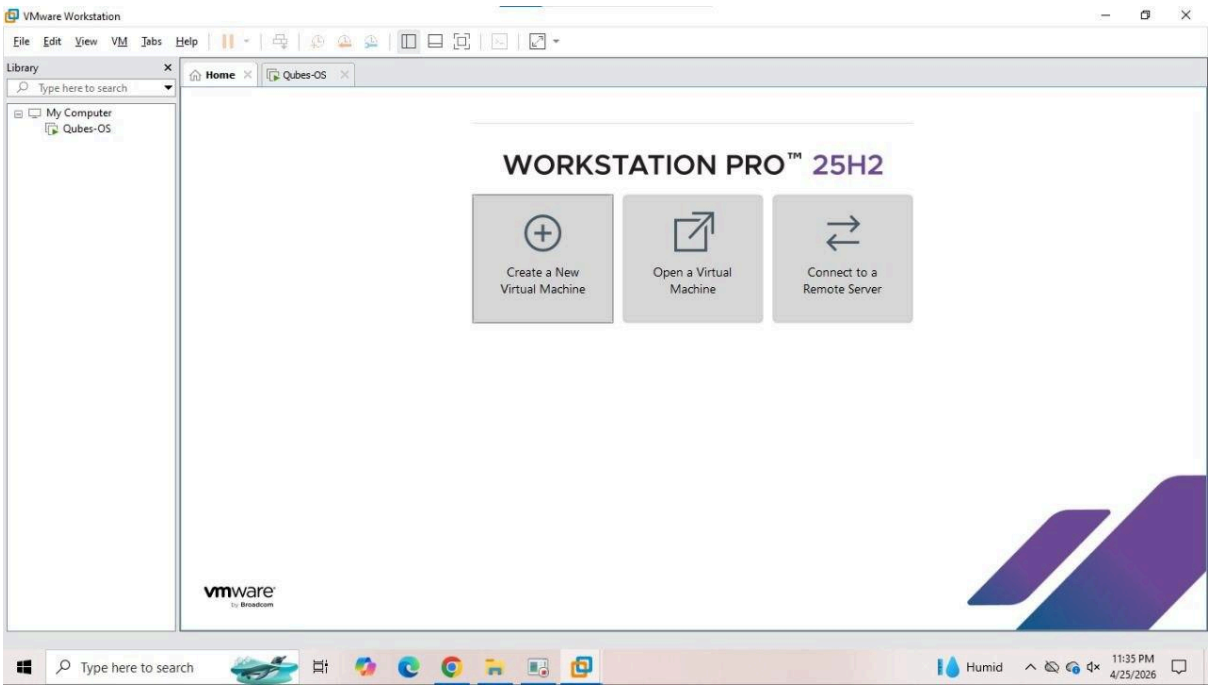

Figure 4: vmware workstation interface

Step 3: Select Installer

- Choose Installer disc image (ISO)
- Browse and select Qubes OS ISO

Step 4: Guest OS Selection

- Select Linux → Other Linux (64-bit)

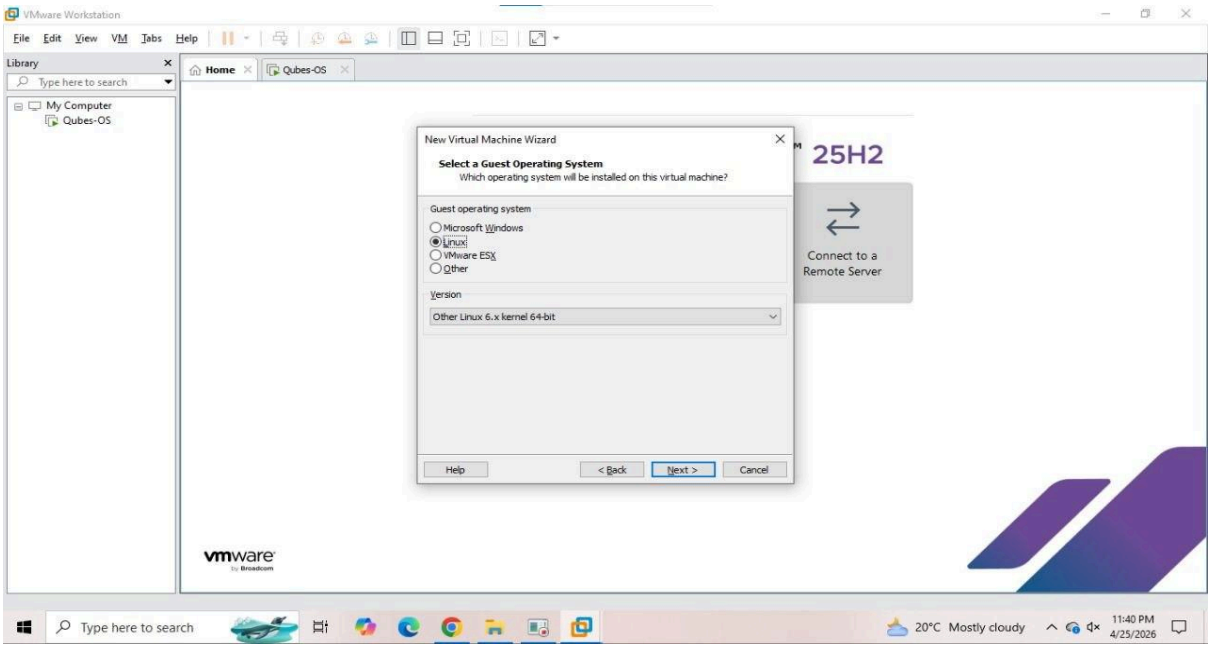

Figure 5: Guest operating system selection

Step 5: Name VM

- Name the VM Qubes OS

Step 6: Allocate Resources

- RAM: 4GB
- CPU: 2 cores
- Disk: 70GB

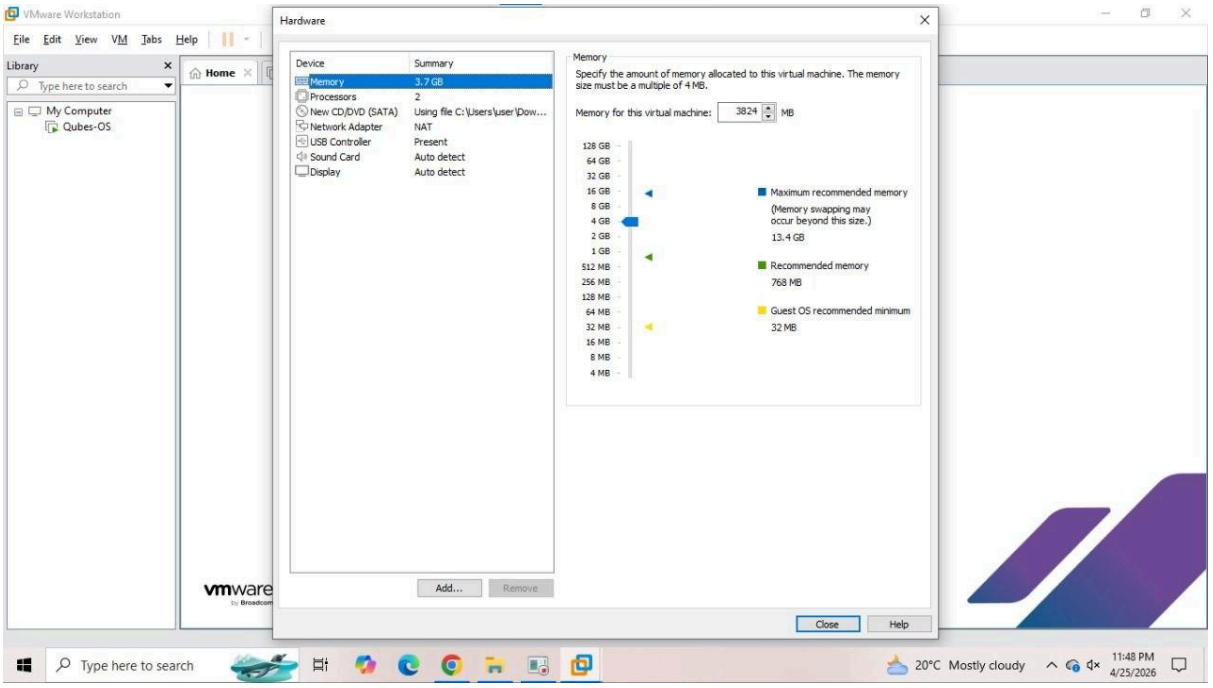

Figure 6: Resource allocation

Step 7: Customize Hardware

- Enable virtualization engine

Step 8: Finish Setup

- Click Finish

1.4.4 Installing Qubes OS

Step 1: Start Virtual Machine

- Click Power On this VM

Step 2: Boot Menu

- Select Install Qubes OS from the boot menu

Step 3: Language Selection

- Choose language(English)
- Click Continue

Step 4: Installation Destination

- Select virtual disk
- Use automatic partitioning

Step 5: Disc Encryption

- ➢ Enable disk encryption
- ➢ Enter passphrase

Step 6: Create User Account

- Enter full name: Beletu Dires
- Set password

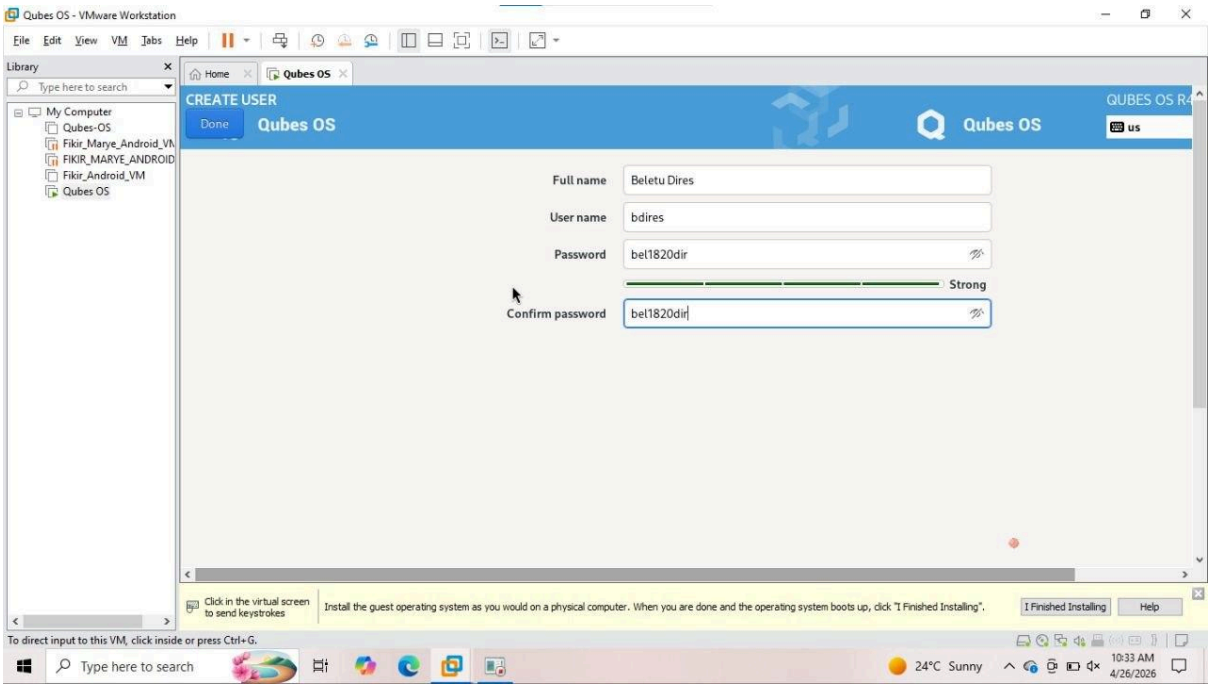

Figure 7: User account creation

Step 5: Begin Installation

- Click Begin Installation

Step 8: Reboot

- Click Reboot System

Step 9: Initial Setup

- Enter the disk encryption passphrase
- The "Initial Setup" screen allows for template configuration
- Click "Finish Configuration to finalize the environment setup
- Log in using the user credentials created during installation
- The system boots into "domo," which is the main administrative qube that manages the desktop environment

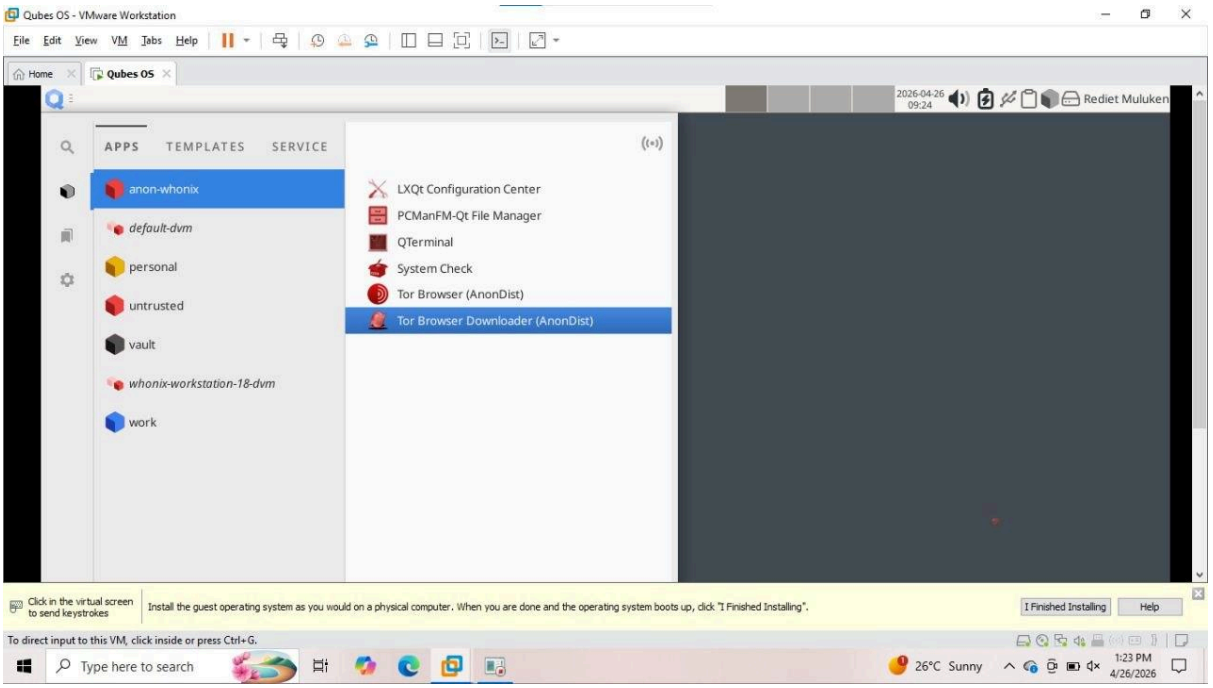

Figure 8: Qubes interface

1.5. Issues (Problems Faced)

- **Issue 1: Virtualization Not Enabled:** The virtual machine failed to start due to disabled virtualization support.

Cause: Hardware virtualization (VT-x/AMD-V) is disabled in BIOS/UEFI.

- **Issue 2: Slow installation speed**: The system becomes slow during installation.

Cause: Insufficient RAM Allocation- Qubes OS requires high memory resources.

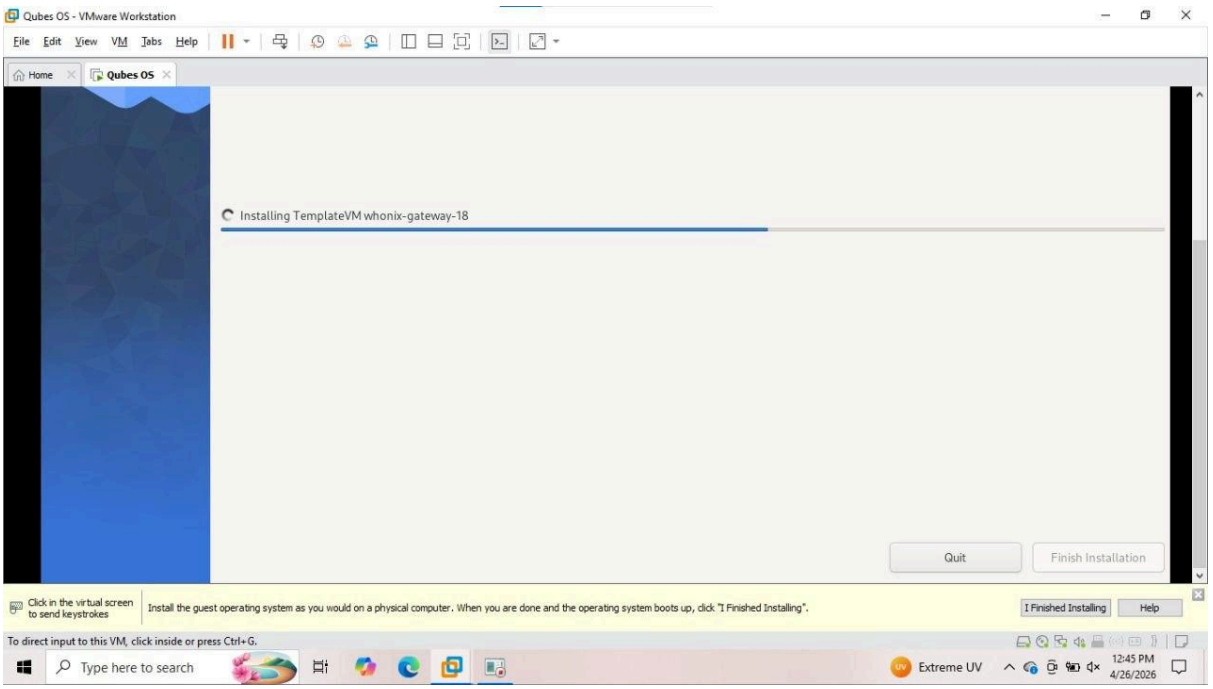

Figure 9: Slow speed qubes installation

- **Issue 3: Slow Performance After Installation:** The system is laggy after installation.

Cause: Limited hardware resources.

1.6. Solutions and Workarounds

Issue	Solution
Issue 1	Enable VT-x/AMD-V in BIOS
Issue 2	Allocate at least 8GB RAM
Issue 7	Increase RAM/CPU allocation

1.7. Filesystem Support

Qubes OS, being a Linux-based operating system, supports a variety of filesystems commonly found in the Linux ecosystem. The primary filesystems used and supported by Qubes OS are ext4 and Btrfs. Understanding these choices and comparing them with other common filesystems like NTFS and FAT32 is essential.

Filesystems Supported by Qubes OS

- **ext4 (Fourth Extended Filesystem)**: This is the default and most widely used filesystem in many Linux distributions, including Qubes OS. ext4 is a journaling

filesystem, offering good performance, reliability, and robustness. It is a mature and well-tested filesystem, making it a safe and stable choice for the root filesystem and user data in Qubes OS.

- **Btrfs (B-tree Filesystem)**: Qubes OS also supports Btrfs, an advanced copy-on-write (CoW) filesystem. Btrfs offers several modern features that are particularly beneficial in a virtualized environment like Qubes OS, such as:
    - **Snapshots**: Efficiently create read-only or writable snapshots of subvolumes, which can be invaluable for system recovery or testing configurations within qubes.
    - **Checksums**: Data and metadata checksums help detect and prevent silent data corruption.
    - **Compression**: Native support for data compression can save disk space.
    - **Subvolumes**: Flexible partitioning and management of filesystem structures, which can be useful for managing different qubes.
      The CoW nature ofBtrfs means that when data is modified, new blocks are written, and the old blocks are retained until no longer referenced, which can be advantageous for VM management and efficient storage of template VMs.

Why These Are Used

ext4 and Btrfs are chosen for Qubes OS due to their robustness, performance, and feature sets that align with the operating system's design principles. ext4 provides a solid, reliable foundation, while Btrfs offers advanced features that can enhance data integrity, recovery options, and efficient storage management, especially relevant for an OS that heavily relies on virtual machines and their associated data.

Comparison with NTFS and FAT32

Feature	ext4	Btrfs	NTFS	FAT32
Operating System	Linux (Qubes OS)	Linux (Qubes OS)	Windows	Cross-platform (legacy)
Journaling	Yes	Yes (metadata)	Yes	No
CoW	No	Yes	No	No
Snapshots	No (requires LVM)	Yes	No (requires VSS)	No

Feature	ext4	Btrfs	NTFS	FAT32
Checksums	No	Yes (data & metadata)	No	No
Permissions	POSIX (Linux)	POSIX (Linux)	ACL (Windows)	Limited
Max File Size	16 TB	16 EB	16 EB	4 GB
Max Volume Size	1 EB	16 EB	256 TB	2 TB
Use Case	General-purpose Linux, system partitions	Advanced Linux, VM storage, data integrity	Windows system, large files	Removable media, small files

NTFS (New Technology File System) is the primary filesystem for Windows operating systems. While it offers features like journaling, security descriptors (ACLs), and support for large files and volumes, it lacks native support for Linux-specific permissions and advanced features like CoW and snapshots that are beneficial for Qubes OS. Using NTFS for Qubes OS system partitions would be highly impractical and lead to significant compatibility and security issues.

FAT32 (File Allocation Table 32) is an older, simpler filesystem widely used for removable media due to its broad compatibility across different operating systems. However, it has severe limitations, including a maximum file size of 4GB and a maximum volume size of 2TB. It lacks journaling, advanced permissions, and any modern data integrity features, making it completely unsuitable for a complex operating system like Qubes OS.

1.8. Advantages and Disadvantages of Qubes OS

Advantages

- **Strong Security Through Isolation**: Qubes OS's primary advantage is its robust security model, achieved through strict isolation of applications and tasks in separate virtual machines (qubes). This compartmentalization significantly reduces the attack surface, as a compromise in one qube does not necessarily affect others or the sensitive dom0.

- **Uses Xen Hypervisor**: The reliance on the Xen Type-1 bare-metal hypervisor provides a strong foundation for security. Xen's design allows it to manage hardware resources directly, ensuring that the virtualization layer itself is minimal and less prone to vulnerabilities compared to Type-2 hypervisors.
- **Compartmentalization of Tasks**: Users can dedicate specific qubes for different trust levels or tasks (E.g: a "work" qube, a "personal" qube, a "banking" qube, an "untrusted" qube). This prevents cross-contamination of data and activities, enhancing overall privacy and security.
- **Template-Based Management**: Qubes OS uses template VMs, allowing for efficient management and updates. Multiple AppVMs can be based on a single template, saving disk space and simplifying software management.

Disadvantages

- **Not Compatible with VMware Properly**: As demonstrated in this assignment, Qubes OS is not designed to run effectively within Type-2 hypervisors like VMware Workstation. Its dependence on direct hardware access and IOMMU for security features makes nested virtualization a significant challenge, often leading to installation failures or severely degraded performance.
- **High System Requirements**: Due to its architecture, which involves running multiple virtual machines simultaneously, Qubes OS has high hardware requirements, particularly for RAM (8GB+ recommended 16GB) and CPU with virtualization extensions. This can make it inaccessible for users with older or less powerful hardware.
- **Complex Installation and Learning Curve**: The installation process can be more complex than traditional operating systems, especially for users unfamiliar with virtualization concepts. Furthermore, adapting to the qube-based workflow and understanding the security model requires a significant learning curve.
- **Performance Overhead**: While isolation provides security, it comes with a performance overhead. Running multiple virtual machines consumes more resources, which can lead to a less responsive system compared to a single-OS setup on the same hardware.

1.9. Conclusion

This assignment provided a practical, challenging, exploration into the installation of Qubes OS within a VMware Workstation virtual environment. The experience underscored the critical importance of understanding an operating system's architectural design and its specific hardware dependencies. While Qubes OS is a powerful tool for security through isolation, its reliance on the Xen Type-1 hypervisor for bare-metal execution proved to be a significant barrier to successful deployment within a Type-2 hypervisor like VMware. The

numerous issues encountered, from boot failures to hardware virtualization conflicts, vividly demonstrated why Qubes OS is not officially supported in such nested environments.

The exercise served as a valuable learning opportunity, highlighting the complexities of nested virtualization and the fundamental differences between Type-1 and Type-2 hypervisors. It reinforced the understanding that while virtualization offers immense flexibility, it also introduces new layers of abstraction and potential incompatibilities that must be carefully considered. The challenges faced in this assignment are not merely technical hurdles but profound lessons in system architecture and the trade-offs between security, performance, and compatibility in modern computing.

1.10. Future Outlook / Recommendations

Based on the insights gained from this assignment, the following recommendations are provided for future endeavors involving Qubes OS and virtualization:

- **Recommend Installing Qubes OS on Real Hardware**: For any user seeking to leverage the full security benefits and optimal performance of Qubes OS, installation on dedicated bare-metal hardware is the unequivocal recommendation. This approach ensures that the Xen hypervisor has direct and unhindered access to all necessary hardware virtualization features, enabling the robust isolation that Qubes OS is designed to provide.
- **Suggest Improvements in Virtualization Compatibility**: While challenging, continued research and development into improving nested virtualization compatibility for Type-1 hypervisors within Type-2 environments would be beneficial. This could involve enhanced emulation of IOMMU and other critical hardware features by virtualization software like VMware, or more flexible detection mechanisms within Xen to adapt to nested scenarios. However, it is crucial that such improvements do not compromise the fundamental security principles of Qubes OS.
- **Advice for Students Attempting Similar Tasks**: Students undertaking similar assignments should be prepared for significant technical challenges when attempting to run bare-metal hypervisor-based operating systems in nested virtualization. It is essential to:
    - Thoroughly research the architectural requirements of the target OS.
    - Understand the capabilities and limitations of the chosen virtualization software.
    - Be prepared to troubleshoot extensively and accept that full functionality may not be achievable.
    - Focus on the learning outcomes derived from understanding the incompatibilities and technical barriers, rather than solely on achieving a fully functional installation.

2. Virtualization in Modern Operating Systems

2.1. Introduction

Virtualization, in the context of modern operating systems, refers to the creation of a virtual (rather than actual) version of something, such as a server, desktop, storage device, network resources, or even an operating system itself . This technology allows a single physical hardware system to host multiple isolated virtual environments, each capable of running its own operating system and applications independently. The importance of virtualization in today's computing landscape cannot be overstated, profoundly impacting areas such as cloud computing, cybersecurity, and software development. It serves as a foundational technology for efficient resource utilization, enhanced security postures, and flexible development workflows, driving innovation and operational efficiency across various industries .

2.2. What is Virtualization?

At its core, virtualization is a technology that abstracts computing resources from the underlying physical hardware. This abstraction layer enables the creation of virtual representations of physical components, allowing them to be shared among multiple virtual entities. The primary components facilitating this are **virtual machines (VMs)** and **hypervisors**.

A **Virtual Machine (VM)** is a software-based emulation of a physical computer. It functions as a complete, isolated computing environment, encompassing its own virtual CPU, memory, storage, and network interfaces. Each VM runs an independent operating system, known as the **guest OS**, which operates as if it were on dedicated hardware. This isolation ensures that applications running within one VM do not interfere with those in another, even when sharing the same physical host.

A **Hypervisor**, also known as a Virtual Machine Monitor (VMM), is the crucial software layer that creates and manages VMs. It acts as an intermediary between the physical hardware and the virtual machines, allocating physical resources to each VM and ensuring their isolated operation. Hypervisors are categorized into two main types:

- **Type 1 Hypervisors (Bare-Metal Hypervisors):** These hypervisors run directly on the host hardware, without an underlying operating system. They have direct access to the hardware resources, offering superior performance, security, and stability. Examples include VMware ESXi, Microsoft Hyper-V, and Xen.
- **Type 2 Hypervisors (Hosted Hypervisors):** These hypervisors run as an application on a conventional operating system (the host OS). They rely on the host OS for hardware access and resource management. While offering greater ease of installation and use, they typically incur more overhead and have lower performance compared to

Type 1 hypervisors. Examples include Oracle VirtualBox, VMware Workstation, and Parallels Desktop.

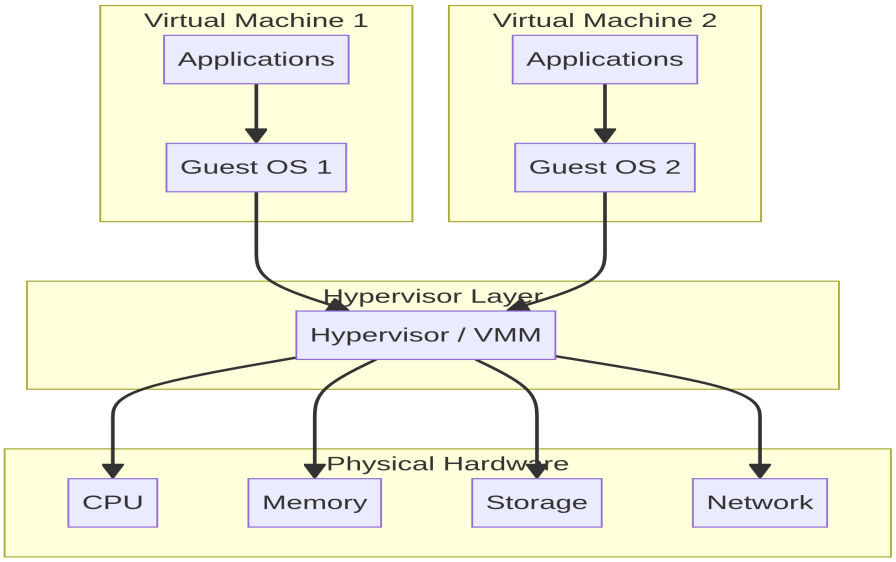

Figure 10: Virtualization Architecture

2.3. Why Virtualization is Important

Virtualization has become indispensable in modern computing due to its numerous benefits:

- **Resource Optimization:** Virtualization allows for the consolidation of multiple workloads onto fewer physical servers. By running several VMs on a single physical machine, organizations can significantly increase hardware utilization, reducing the need for additional physical infrastructure. This leads to a more efficient use of CPU, memory, storage, and network resources, minimizing idle capacity.
- **Isolation and Security:** Each virtual machine operates in an isolated environment, meaning that issues or security breaches within one VM do not affect others on the same physical host. This isolation provides a robust security boundary, making it an ideal solution for running applications with different security requirements or for containing potential threats.
- **Scalability and Flexibility:** Virtualized environments offer unparalleled scalability. New VMs can be provisioned rapidly to meet increasing demand, and resources can be dynamically allocated or deallocated as needed. This flexibility allows businesses to quickly adapt to changing workloads and expand their infrastructure without significant hardware investments.
- **Cost Efficiency:** By optimizing resource utilization and reducing the number of physical servers, virtualization directly translates into significant cost savings. These savings stem from lower hardware procurement costs, reduced power consumption, decreased cooling requirements, and simplified management, leading to a lower total cost of ownership.

- **Development and Testing Benefits:** Virtualization provides developers and testers with isolated environments to build, test, and deploy applications without impacting production systems. It allows for the creation of multiple testing environments, easy replication of bugs, and rapid deployment of development sandboxes, accelerating the software development lifecycle.

2.4. How Virtualization Works

The operational mechanics of virtualization revolve around the hypervisor, which orchestrates the entire virtual environment. Here's a step-by-step explanation:

- **Role of the Hypervisor:** The hypervisor is the foundational layer that sits between the physical hardware and the virtual machines. Its primary role is to abstract the physical resources (CPU, memory, storage, network) and present them as virtual resources to each VM. It manages the scheduling of VM access to these physical resources, ensuring fair allocation and preventing conflicts.
- **Creation and Management of Virtual Machines:** When a new VM is created, the hypervisor allocates a portion of the physical resources to it. This includes defining the virtual CPU cores, RAM, disk space, and network interfaces. The hypervisor then loads the guest operating system image into this virtual environment, allowing it to boot and operate as if it were on a dedicated physical machine.
- **Resource Allocation (CPU, Memory, Storage, I/O):** The hypervisor dynamically manages resource allocation. For CPU, it uses techniques like CPU scheduling to give each VM a slice of the physical CPU time. For memory, it employs memory management units (MMUs) to map virtual memory addresses used by VMs to physical memory addresses. Storage is often presented as virtual disks, which are files on the host's physical storage. I/O operations are intercepted by the hypervisor and translated to the underlying physical hardware.
- **Guest OS Execution:** The guest OS within a VM executes its instructions. When these instructions require access to hardware resources, they are trapped by the hypervisor. The hypervisor then translates these requests into commands that the physical hardware can understand and execute. The results are then returned to the guest OS, maintaining the illusion of direct hardware access.
- **Hardware-Assisted Virtualization (Intel VT-x, AMD-V):** Modern CPUs include specialized hardware extensions, such as Intel VT-x and AMD-V, which significantly enhance virtualization performance. These extensions provide new CPU instructions that allow the hypervisor to directly execute certain privileged instructions from the guest OS without requiring software-based translation. This reduces the overhead associated with virtualization, making VMs run closer to native performance.

2.5. Applications of Virtualization

Virtualization is a cornerstone technology across various domains:

- **Cloud Computing:** Virtualization is the fundamental technology enabling cloud computing services. Cloud providers leverage virtualization to create and manage vast pools of virtualized resources (servers, storage, networks) that can be provisioned on-demand to customers. This allows for the flexible and scalable delivery of Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).
- **Cybersecurity:** Virtualization plays a critical role in enhancing cybersecurity. It enables the creation of isolated environments for running potentially malicious software (sandboxing), secure browsing, and setting up honeypots to trap attackers. Furthermore, it facilitates the rapid deployment and rollback of secure operating system environments, aiding in incident response and forensic analysis.
- **Software Testing and Development:** Developers and testers utilize virtualization to create consistent and reproducible testing environments. They can easily provision multiple VMs with different operating systems and configurations to test application compatibility, isolate bugs, and ensure software quality across various platforms. This significantly streamlines the development lifecycle and reduces time-to-market.

2.6. Advantages and Limitations

Advantages

- **Efficiency:** Maximizes hardware utilization, reducing physical server sprawl and energy consumption.
- **Flexibility:** Enables rapid provisioning, dynamic resource allocation, and easy migration of VMs.
- **Security:** Provides strong isolation between virtual machines, enhancing overall system security.
- **Disaster Recovery:** Simplifies backup, recovery, and business continuity planning through VM snapshots and replication.
- **Cost Reduction:** Lowers capital and operational expenses by consolidating hardware and reducing power/cooling needs.

Limitations

- **Overhead:** Despite hardware assistance, virtualization introduces some performance overhead compared to bare-metal execution.

- **Complexity:** Managing a virtualized environment can be complex, requiring specialized skills and tools.
- **Resource Contention:** If not properly managed, multiple VMs competing for limited physical resources can lead to performance degradation.
- **Single Point of Failure:** A failure in the physical host or hypervisor can impact all VMs running on it, though high-availability solutions mitigate this risk.
- **Licensing Issues:** Software licensing for virtualized environments can be complex and costly, depending on the vendor.

2.7. Conclusion

Virtualization has transformed the landscape of modern computing, evolving from a niche technology to an indispensable foundation for enterprise IT, cloud computing, and advanced development practices. By abstracting physical hardware and enabling the creation of isolated virtual environments, it delivers unparalleled benefits in resource optimization, security, scalability, and cost efficiency. While challenges such as performance overhead and management complexity exist, ongoing advancements, particularly in hardware-assisted virtualization, continue to refine its capabilities and broaden its applicability. As computing demands continue to grow, virtualization will undoubtedly remain a critical technology, driving innovation and efficiency across all facets of the digital world.

References

Qubes OS Documentation: System Requirements. (n.d.). Retrieved from https://doc.qubes-os.org/en/latest/user/hardware/system-requirements.html

Qubes OS Documentation: Architecture. (n.d.). Retrieved from https://doc.qubes-os.org/en/latest/developer/system/architecture.html

Qubes OS Forum: QubesOS in VMWare Workstation 17 Pro. (n.d.). Retrieved from https://forum.qubes-os.org/t/qubesos-in-vmware-workstation-17-pro/32628

Qubes OS Forum: Can I install QubesOS in a Qemu KVM Virtual machine?. (n.d.). Retrieved from https://forum.qubes-os.org/t/can-i-install-qubesos-in-a-qemu-kvm-virtual-machine/30763

Reddit: Is there a way to run Qubes through a virtual machine? (n.d.). Retrieved from https://www.reddit.com/r/Qubes/comments/ga3iul/cubes_as_a_general_hypervisor/

Kicksecure: Nested Virtualization. (n.d.). Retrieved from https://www.kicksecure.com/wiki/Nested_Virtualization

Qubes OS Forum: Ext4 vs. Btrfs performance on Qubes OS installs. (n.d.). Retrieved from https://forum.qubes-os.org/t/ext4-vs-btrfs-performance-on-qubes-os-installs/13585

Reddit: File System. (n.d.). Retrieved from https://www.reddit.com/r/Qubes/comments/17qv9ri/file_system/

Red Hat. (n.d.). *What is virtualization?* Retrieved from https://www.redhat.com/en/topics/virtualization/what-is-virtualization IBM. (n.d.). *What Is Virtualization?* Retrieved from https://www.ibm.com/think/topics/virtualization AWS. (n.d.). *What is Virtualization? - Cloud Computing ...* Retrieved from https://aws.amazon.com/what-is/virtualization/

Oracle. (n.d.). *What Is Virtualization?* Retrieved from https://www.oracle.com/virtualization/virtualization-explained/

AWS. (n.d.). *Type 1 vs Type 2 Hypervisors - Difference Between ...* Retrieved from https://aws.amazon.com/compare/the-difference-between-type-1-and-type-2-hypervisors/

Stormagic. (2025, May 20). *Type 1 vs Type 2 Hypervisors: Key Differences Explained*. Retrieved from https://stormagic.com/company/blog/type-1-vs-type-2-hypervisors/

DigitalOcean. (2026, February 3). *10 Benefits of Virtualization in Cloud Computing for 2026*. Retrieved from https://www.digitalocean.com/resources/articles/benefits-of-virtualization

Veeam. (2024, January 20). *The Top Benefits of Virtualization for Your Business*. Retrieved from https://www.veeam.com/blog/benefits-of-virtualization.html

NFina. (2026, April 15). *Maximizing Efficiency: Exploring the Benefits of Virtualization*. Retrieved from https://nfina.com/benefits-of-virtualization/

IBM. (n.d.). *The Benefits Of Virtualization*. Retrieved from https://www.ibm.com/think/insights/virtualization-benefits

Pass4Sure. (n.d.). *Understanding Virtualization and Its Importance in Modern ...* Retrieved from https://www.pass4sure.com/blog/understanding-virtualization-and-its-importance-in-modern-computing/

Nutanix. (2025, November 19). *What is Virtualization? The Complete Guide*. Retrieved from https://www.nutanix.com/info/virtualization

StarWind Software. (2026, March 15). *What is Virtualization? A Deep Dive Guide*. Retrieved from https://www.starwindsoftware.com/blog/virtualization/

Red Hat. (2023, January 3). *What is a hypervisor?* Retrieved from https://www.redhat.com/en/topics/virtualization/what-is-a-hypervisor

Invgate. (2024, July 22). *What is a Hypervisor? A Simple Explanation*. Retrieved from https://blog.invgate.com/what-is-a-hypervisor

Wikipedia. (n.d.). *X86 virtualization*. Retrieved from https://en.wikipedia.org/wiki/X86_virtualization

] AWS. (n.d.). *What is a Hypervisor?* Retrieved from https://aws.amazon.com/what-is/hypervisor/

SentinelOne. (2025, July 16). *What is a Hypervisor? Types, Benefits, and Best Practices*. Retrieved from https://www.sentinelone.com/cybersecurity-101/cybersecurity/what-is-a-hypervisor/

GeeksforGeeks. (2025, December 8). *Hypervisor*. Retrieved from https://www.geeksforgeeks.org/system-design/hypervisor/